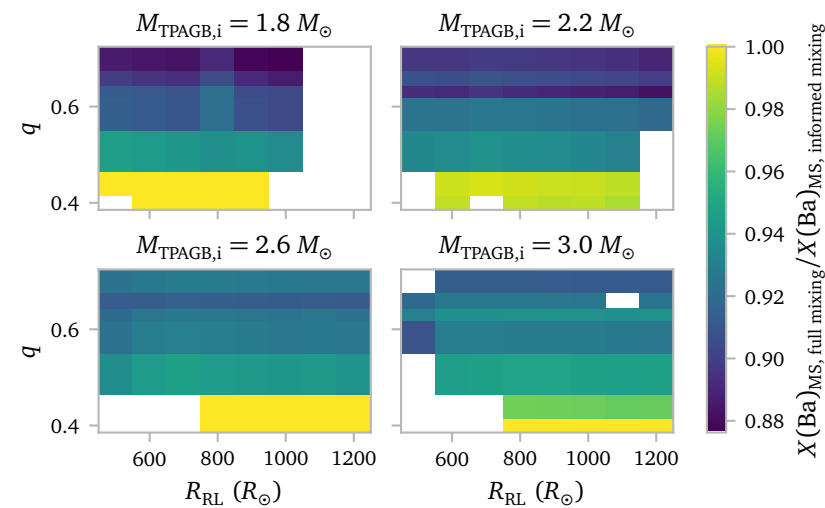


Notes Week 30

a. Checking differences between mixing methods

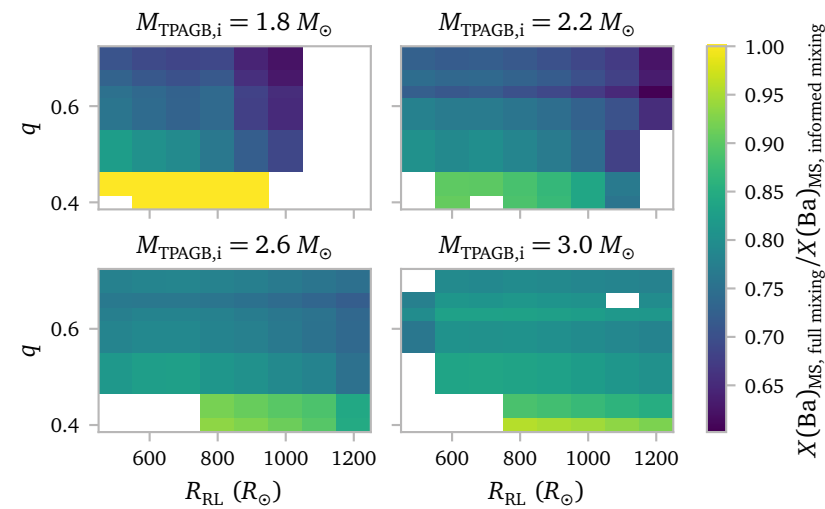
Last week, I showed *this* plot



We would expect larger differences if we have a significant accretor core that is not mixed. Let me investigate.

I quite quickly found the issue, which actually changes the results for every plot I showed last week that was related to the binary abundance post-processing.

I accidentally took the final accretor mass $M_{\text{acc},f}$ as the total amount of accreted mass, while this *obviously* should be $\Delta M_{\text{acc}} = M_{\text{acc},f} - M_{\text{acc},i}$. This is an easy fix and now the plot above looks like this:



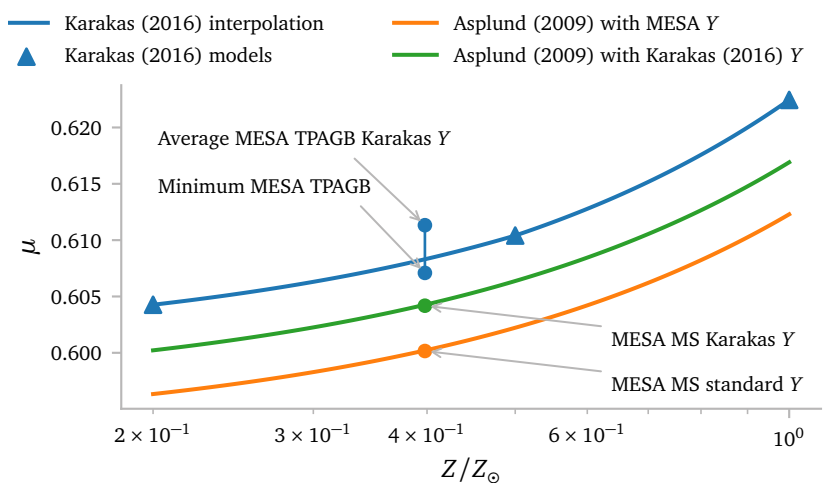
b. Mean Molecular weight MESA vs Monash

We saw that there is a difference between the mean molecular weight of the MESA models and the initial mean molecular weight of the Monash models.

I want to understand where these differences come from.

Let me start by computing μ from the [Asplund et al. \(2009\)](#) solar abundances for the MESA initial Helium mass-fraction.

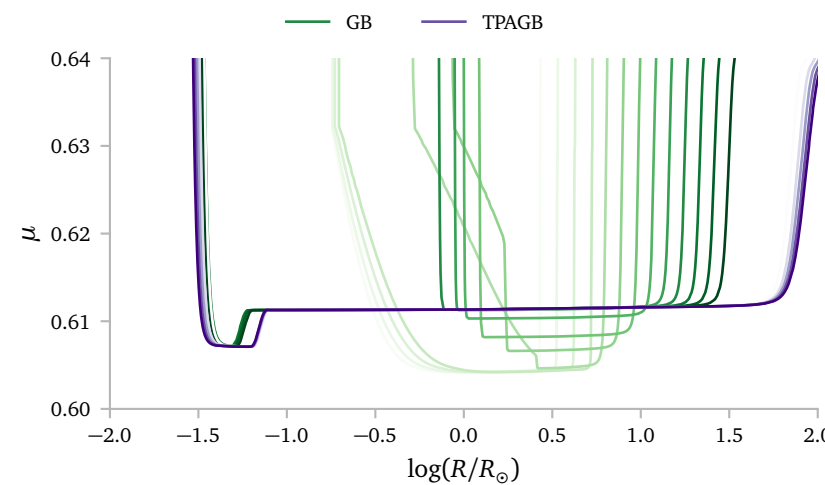
Then I will modify my [Asplund et al. \(2009\)](#) abundances to the “canonical” helium mass-fraction as explained in [Karakas \(2014\)](#).



The mean molecular weights of the MESA models on the main sequence agree very well with the mean molecular weights of the [Asplund et al. \(2009\)](#) for both the standard MESA initial helium mass-fraction as well as the “canonical” [Karakas \(2014\)](#) mass-fraction.

The MESA models do not however, agree that well with the [Karakas & Lugaro \(2016\)](#) mean molecular weights.

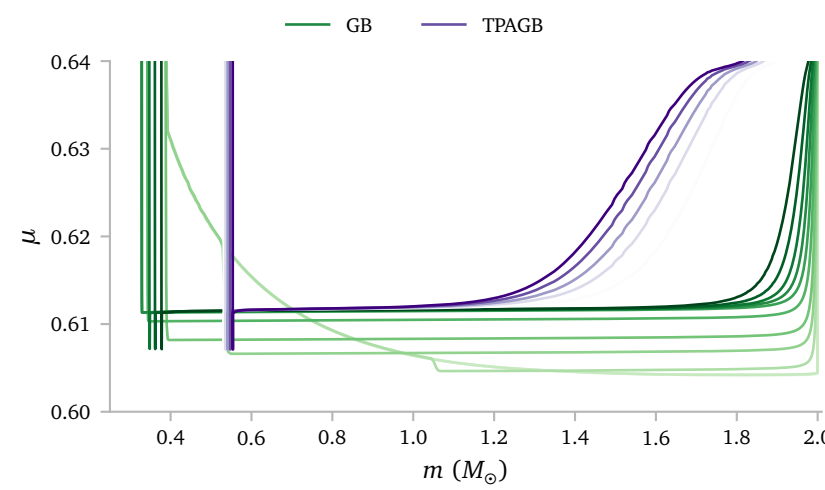
I plotted *two* values for the initial TPAGB MESA model. Below, I show some profiles during the evolution of the MESA model with the [Karakas \(2014\)](#) helium mass-fraction.



There is a little dip just past the helium core boundary. The bottom blue dot in the plot above is sampled from this dip. The upper blue dot is sampled at $R = 1$, which is the value of most of the envelope, and probably the most relevant parameter.

The dip first forms towards the end of the GB already. This is *after* the first dredge-up.

We can obviously also plot it with mass on the horizontal axis. Then it becomes very clear that the small dip is not the relevant value to use.



The minimum is barely visible here. We can also clearly see the growth of the core between the GB and the TPAGB.

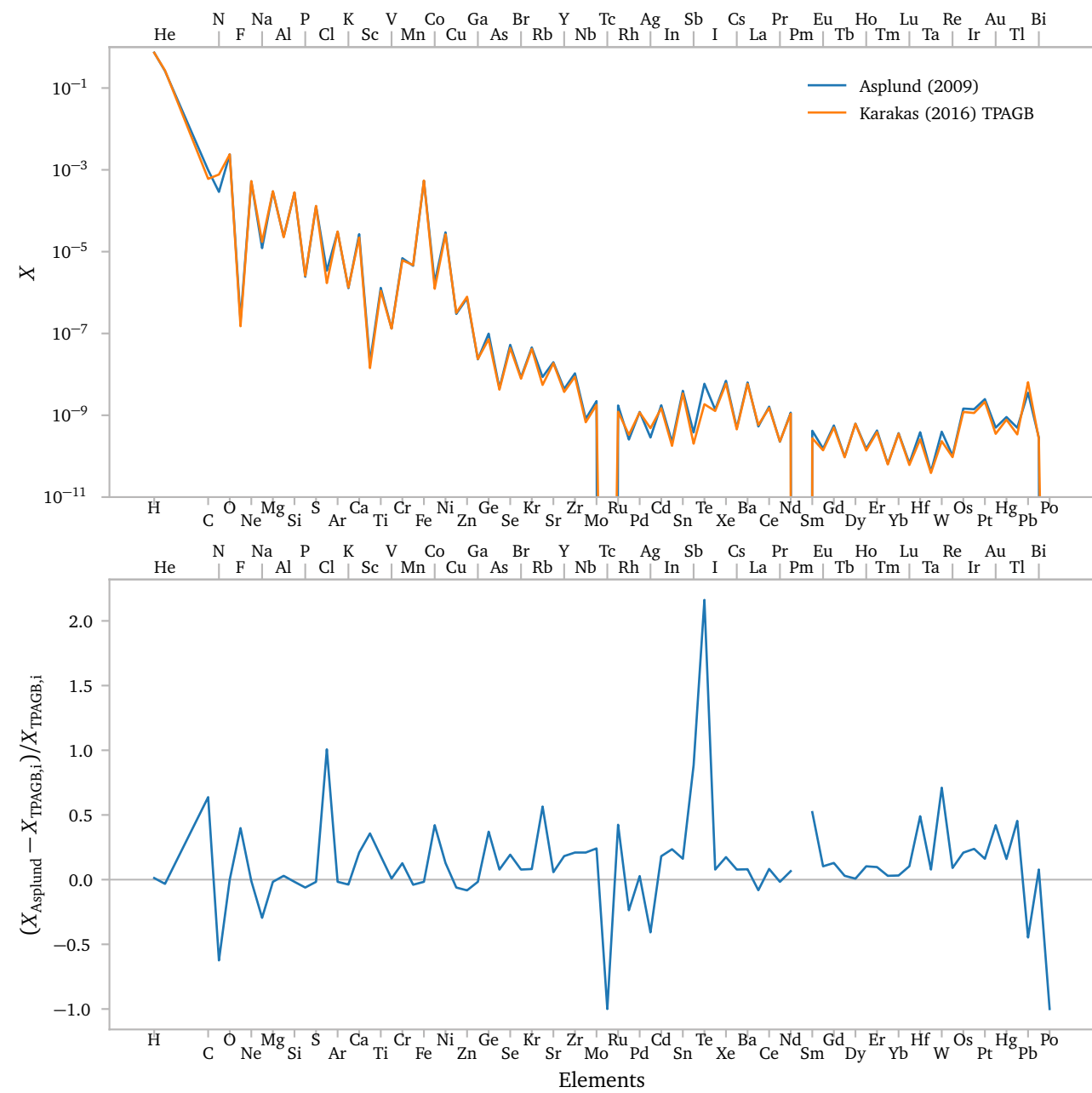
I feel like the best thing to do would be to simulate everything with the [Karakas \(2014\)](#) initial helium mass fraction. This would however mean that I would need to simulate the single stars again, which takes some time and manual help.

An easy first thing to do though is to simulate the accretors with the [Karakas \(2014\)](#) initial helium abundances.

This would at least ensure that the mean molecular weight profile of the main sequence stars should align with the mean molecular weight of the Monash models.

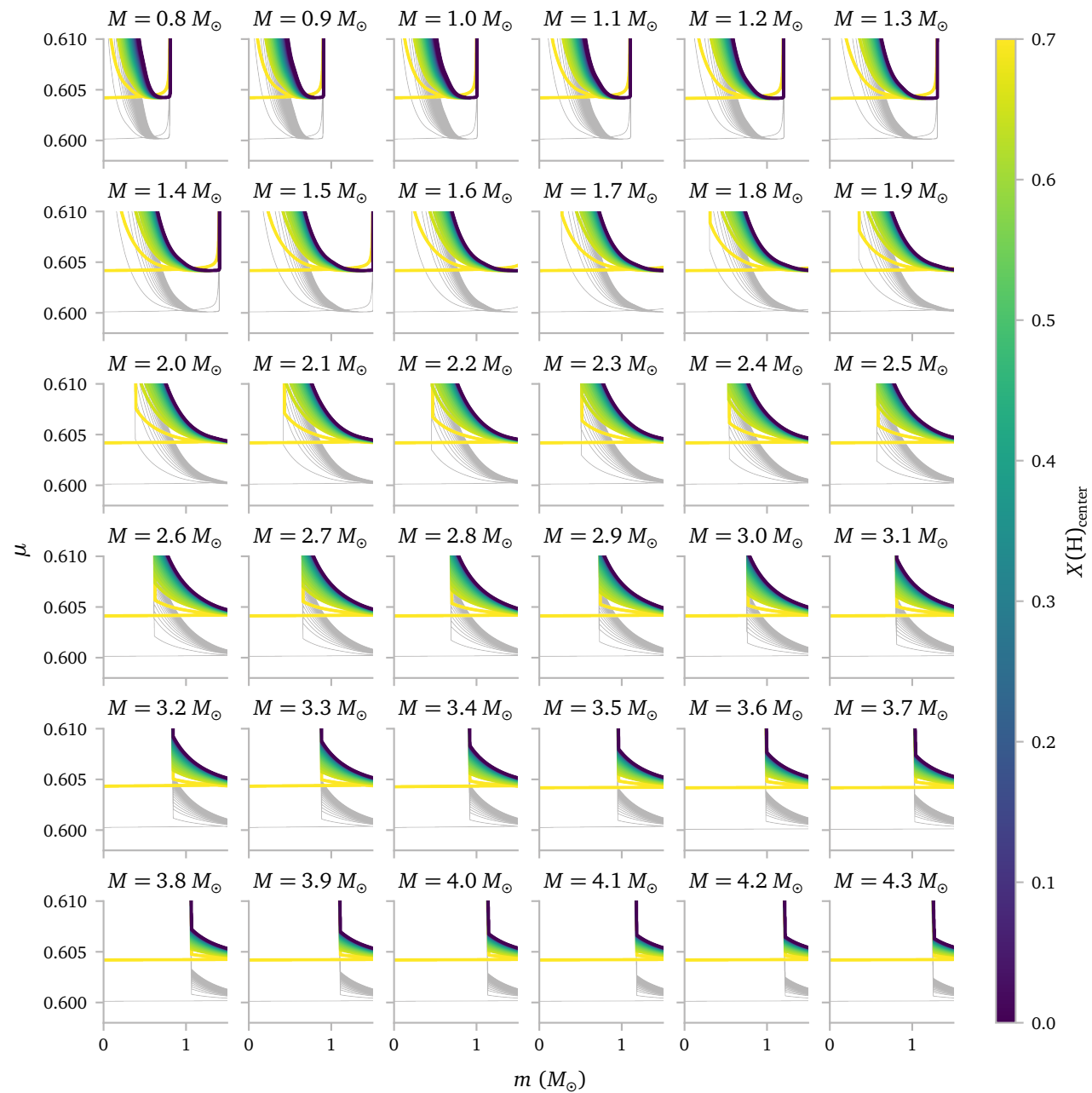
However, the actual MESA binary models would still evolve with a different helium mass fraction. I also think I should use the [Asplund et al. \(2009\)](#) MS abundances for the accretor *instead of*, what I do not, use the initial envelope abundances of the TPAGB stars.

This is quite easy.

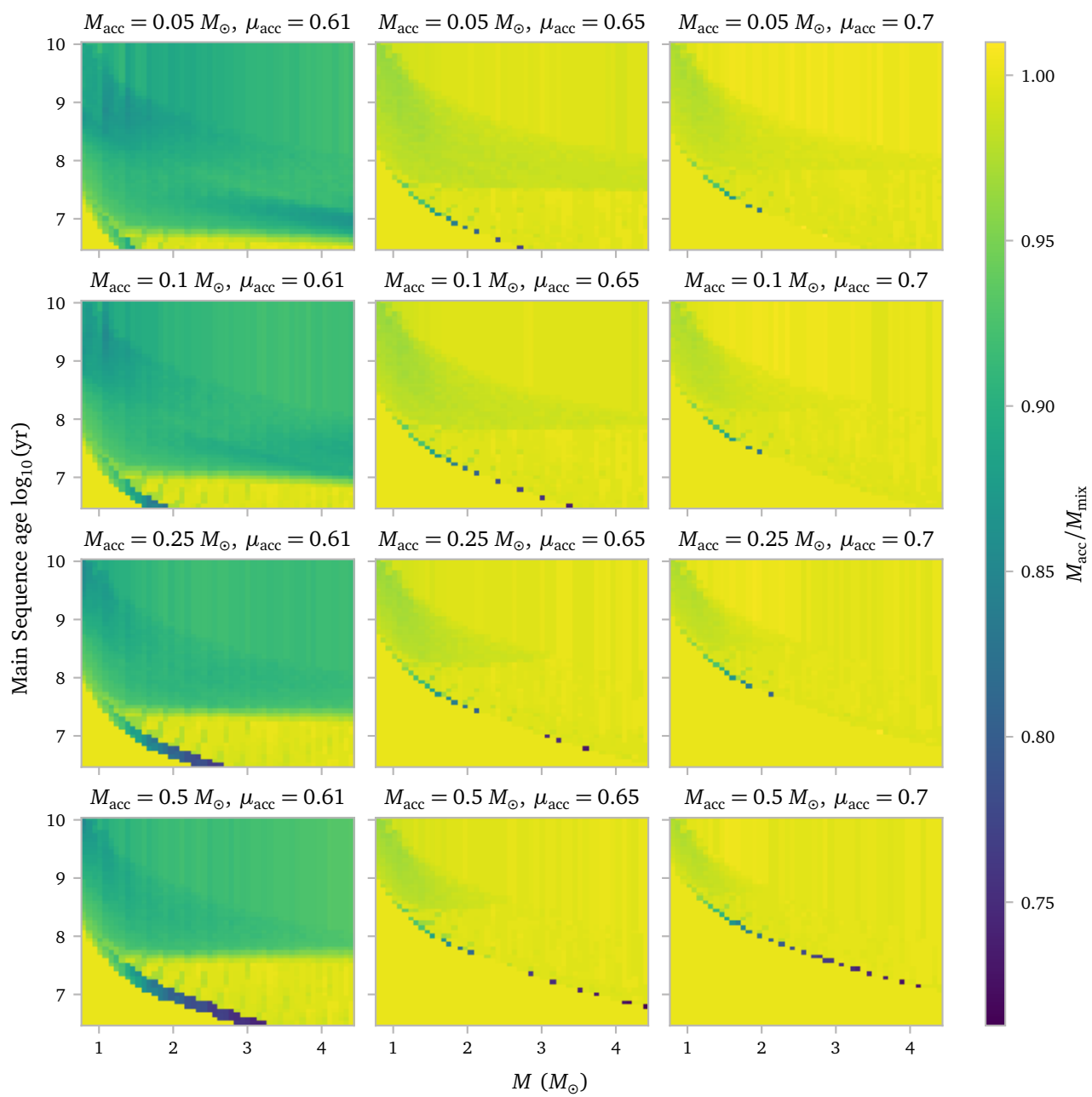


I simulated the main sequence stars with the canonical helium fractions from Karakas (2014).

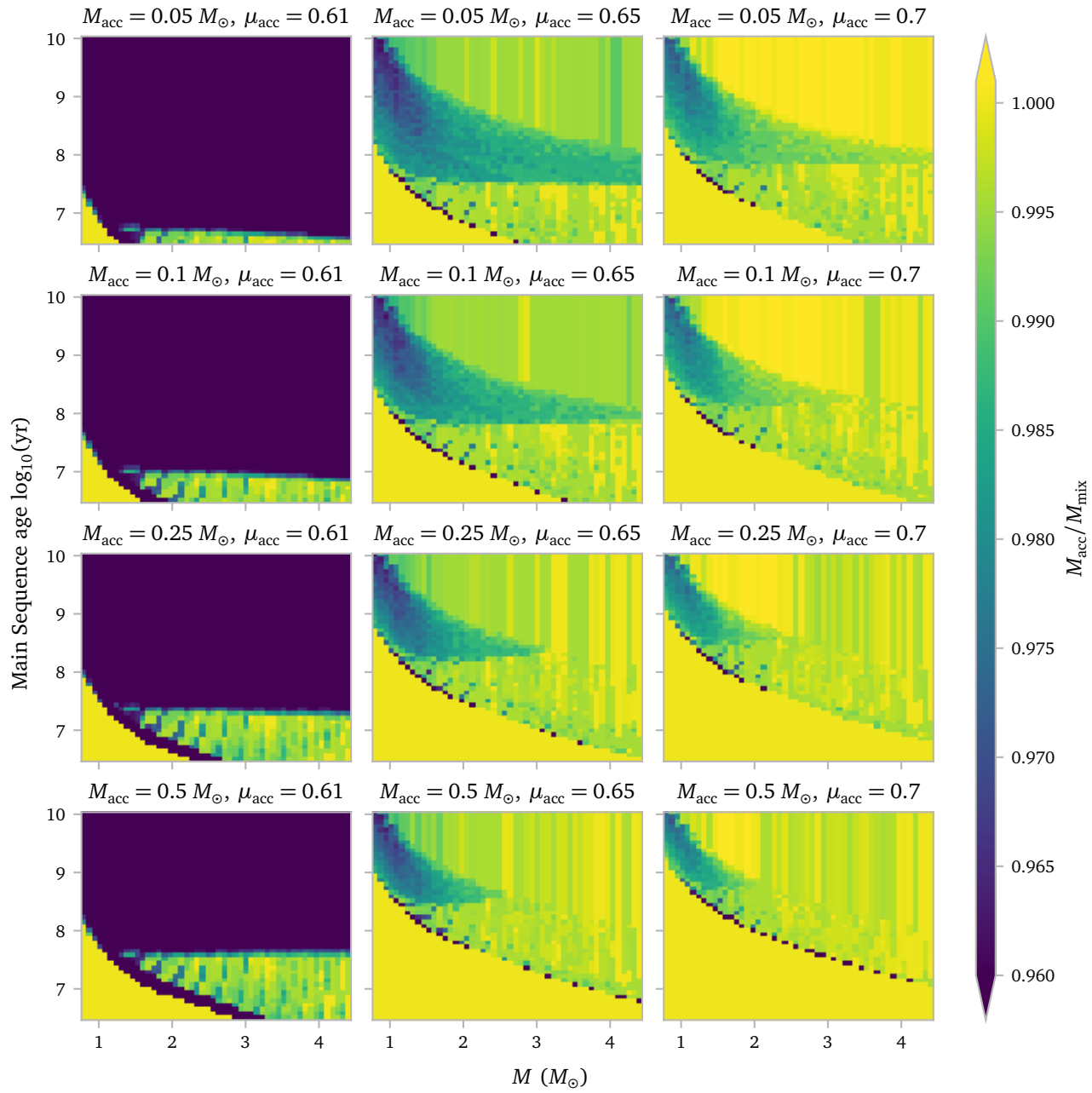
Here is the zoom on the mean molecular weight of the MESA helium mass fractions in gray, and the new ones in color.



The envelope has slightly larger mean molecular weight values. This *should* mean we get slightly lower mixing masses.

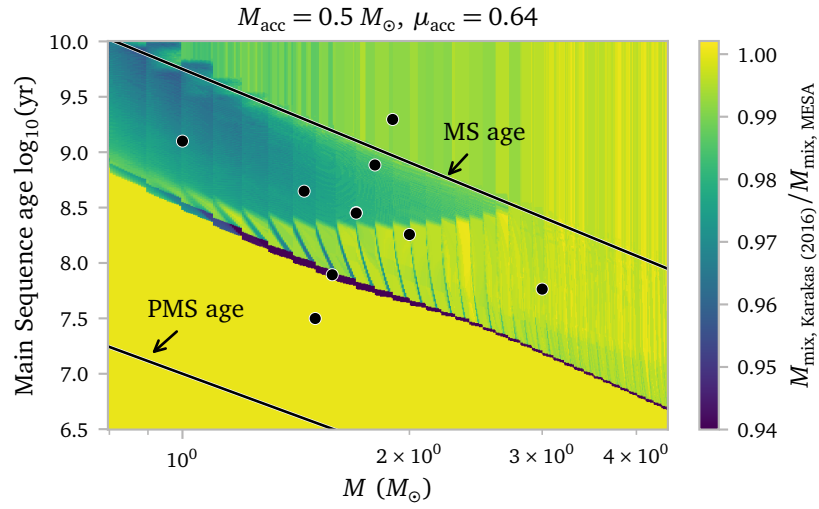


This is indeed what we see. It is much clearer for the second and third column if we zoom in the colors a bit.



However, we also see that only certain regions are affected.

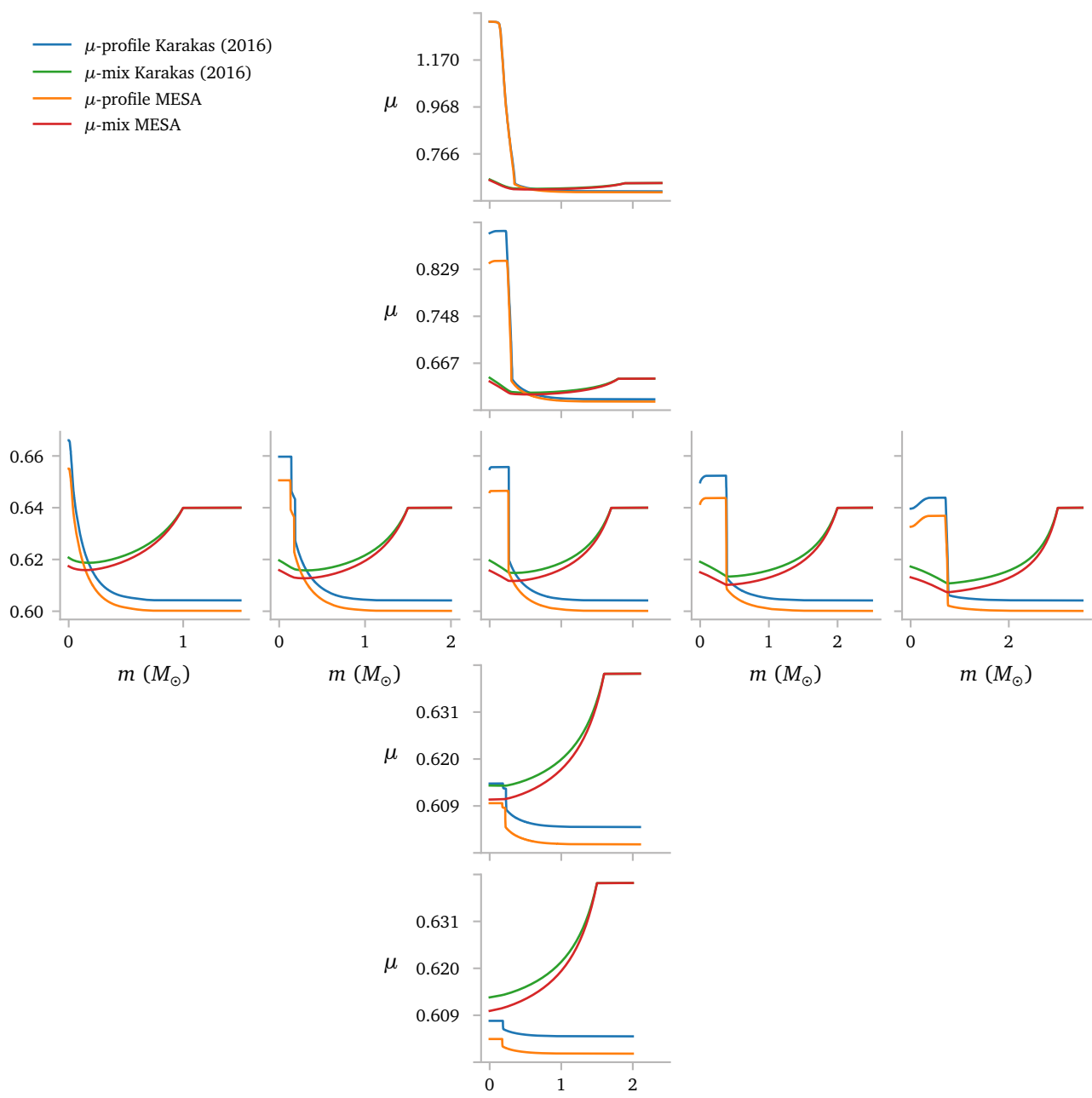
Below, I zoom in on a particular case of M_{acc} and μ_{acc} . Lastly, I



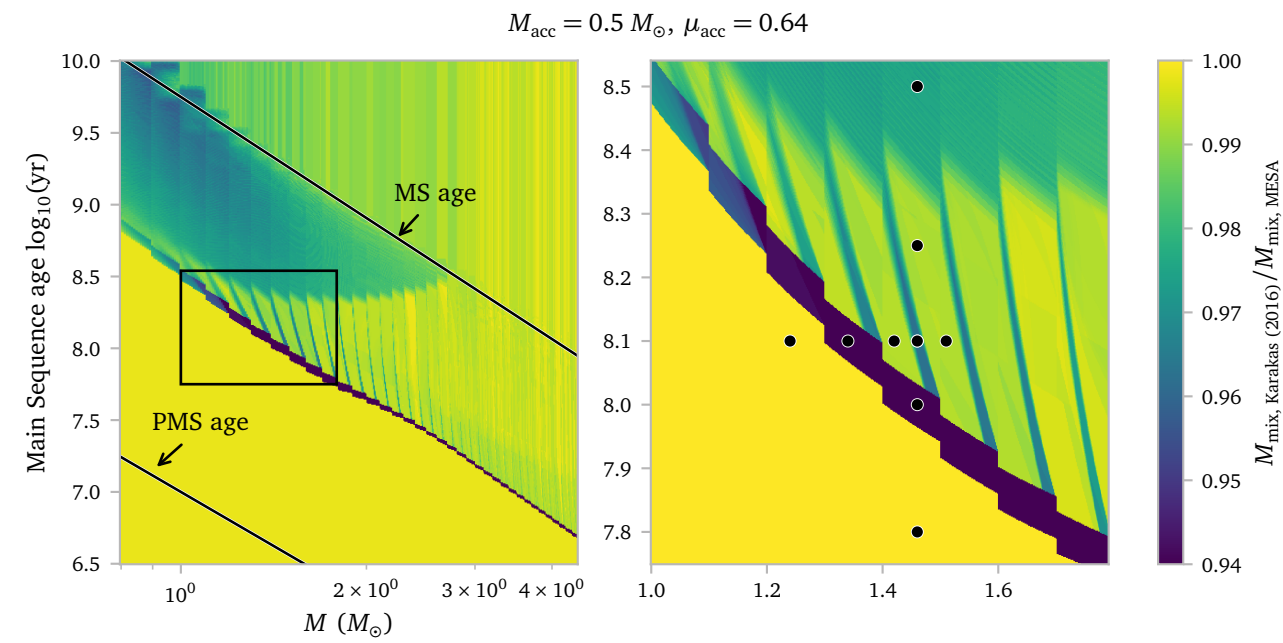
The diagonal lines show the approximate ages for the pre-MS and the MS.

I show the μ -profiles of the cross formation of dots in the figure below.

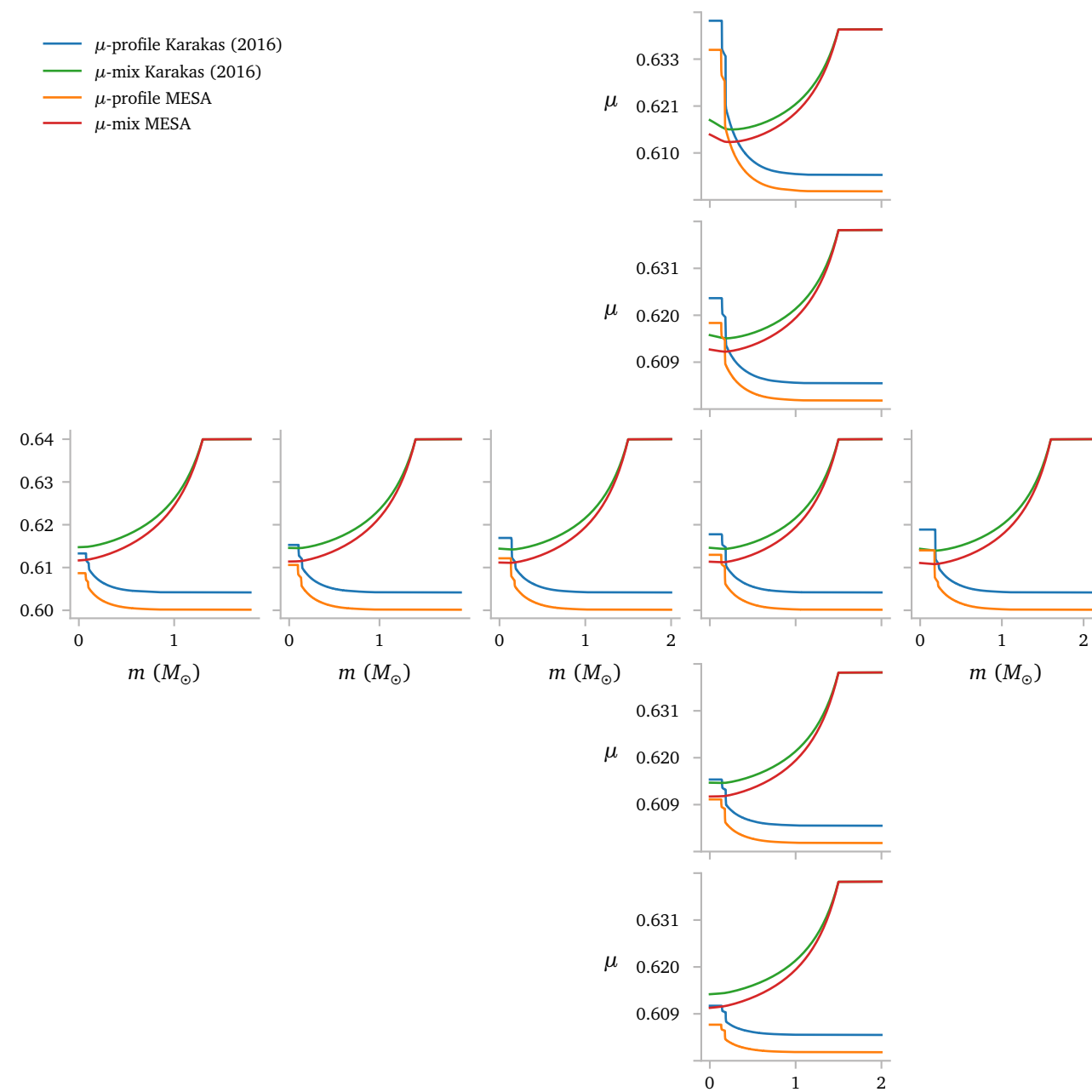
The profiles clearly show where the differences between the two comes from.



There are also these thin lines. I zoom in one those below.

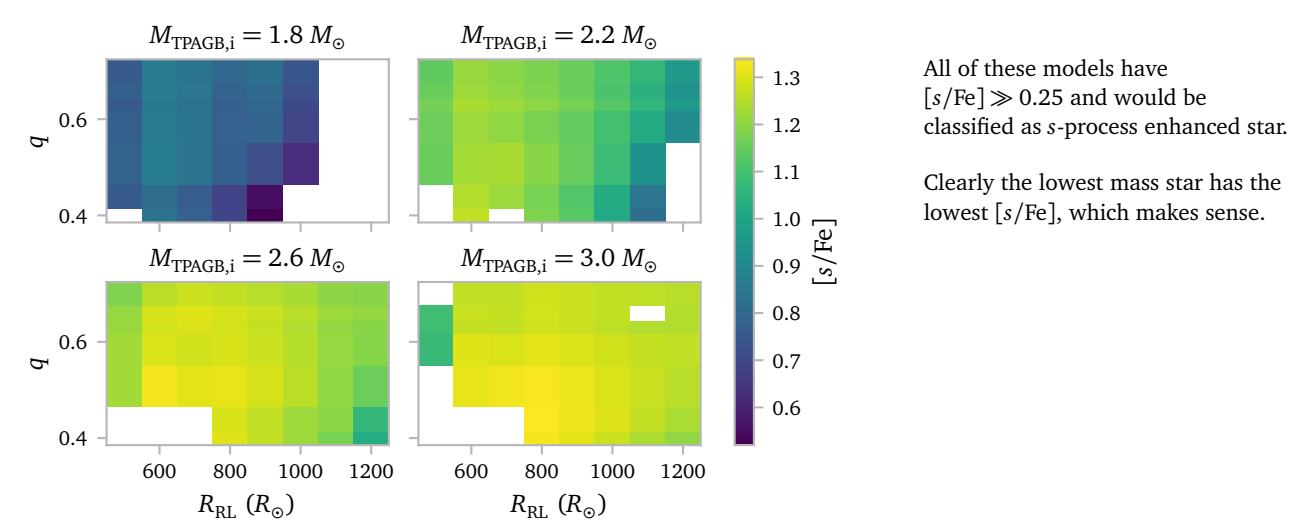
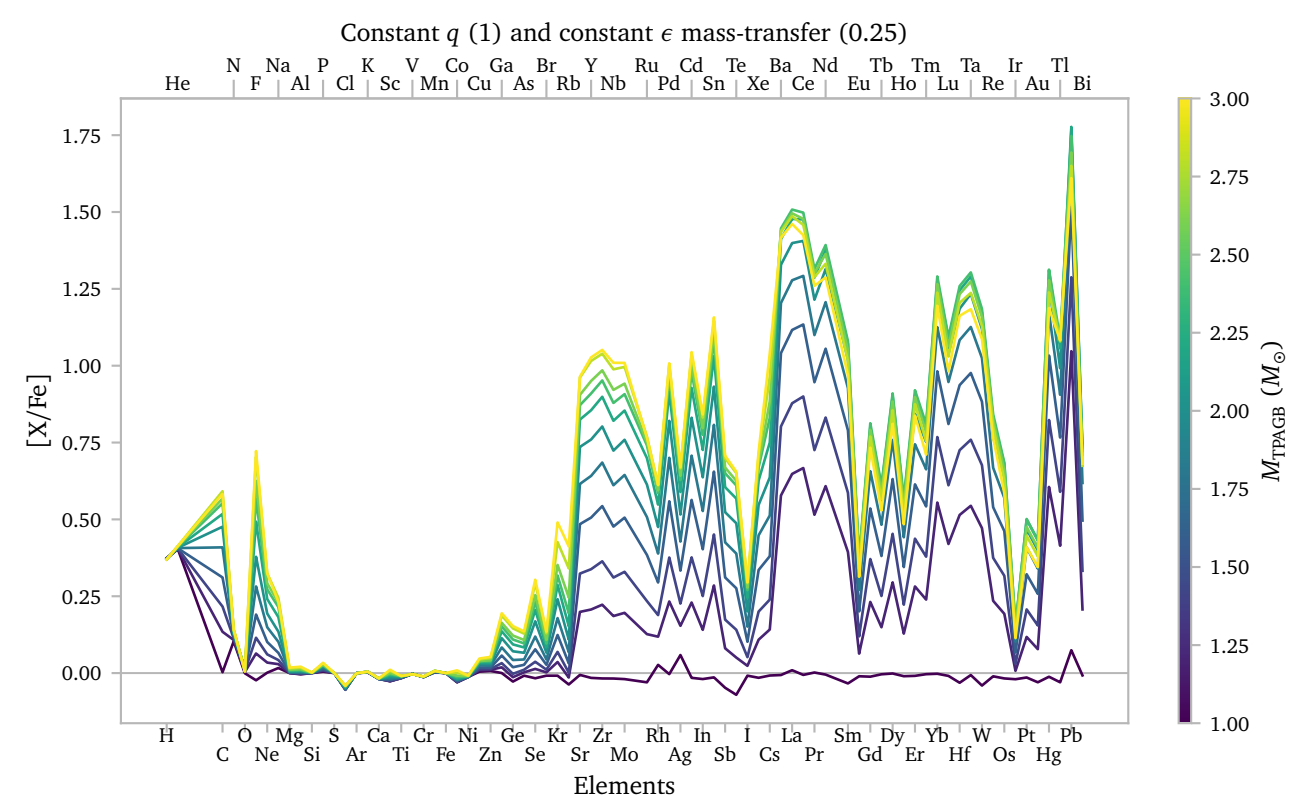
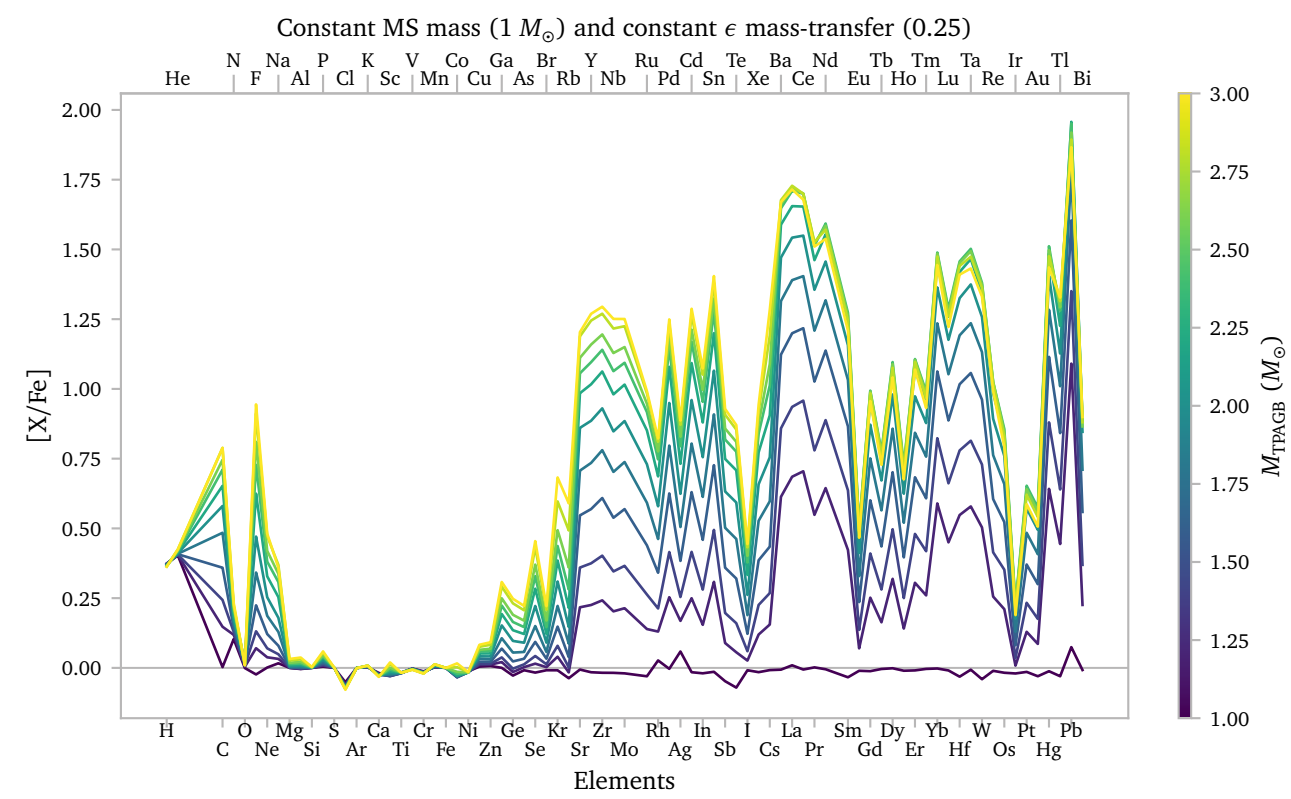
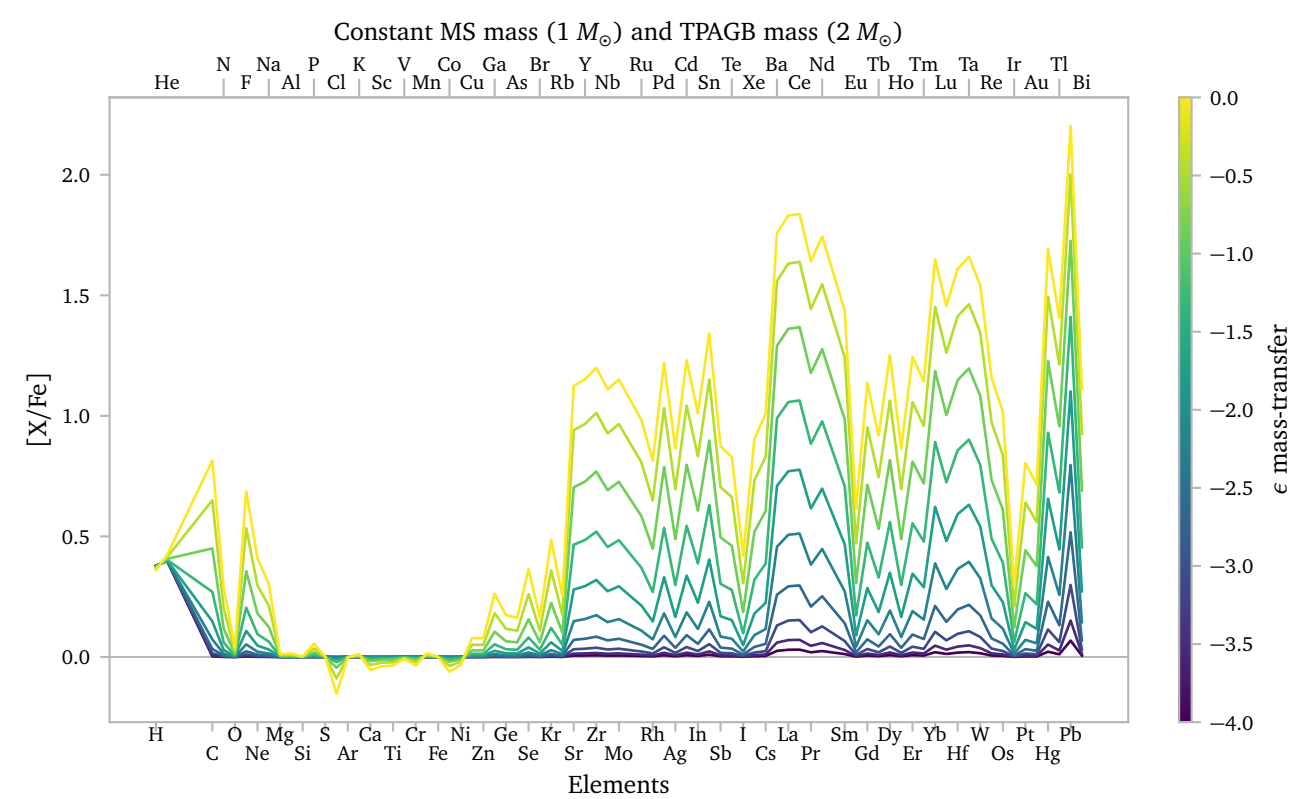


Again, the dots from a cross formation and the profiles are shown below.



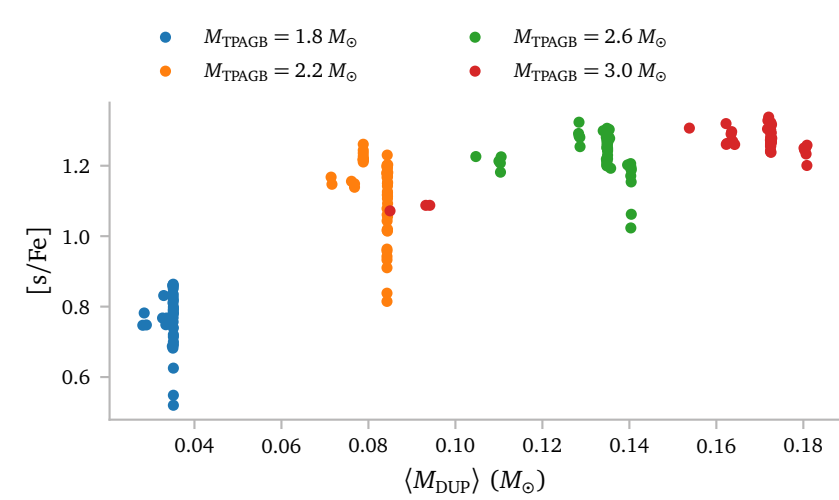
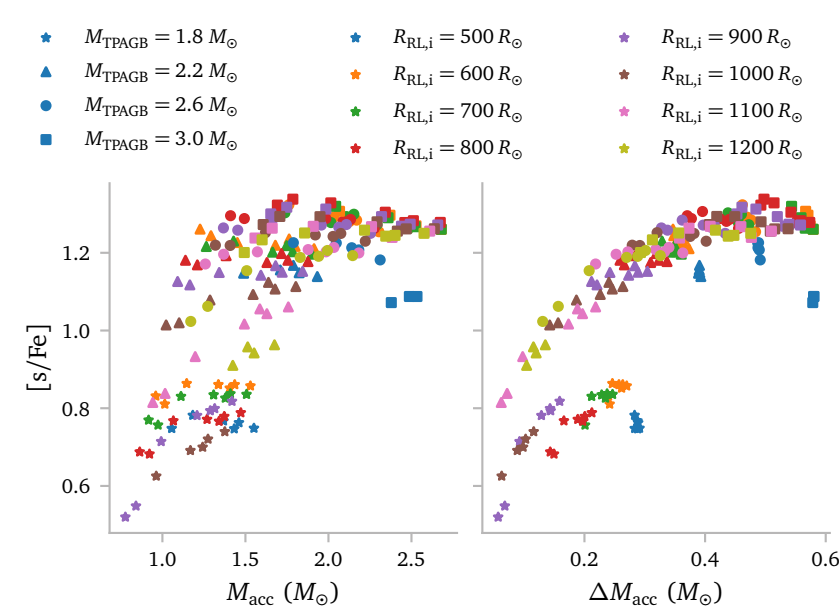
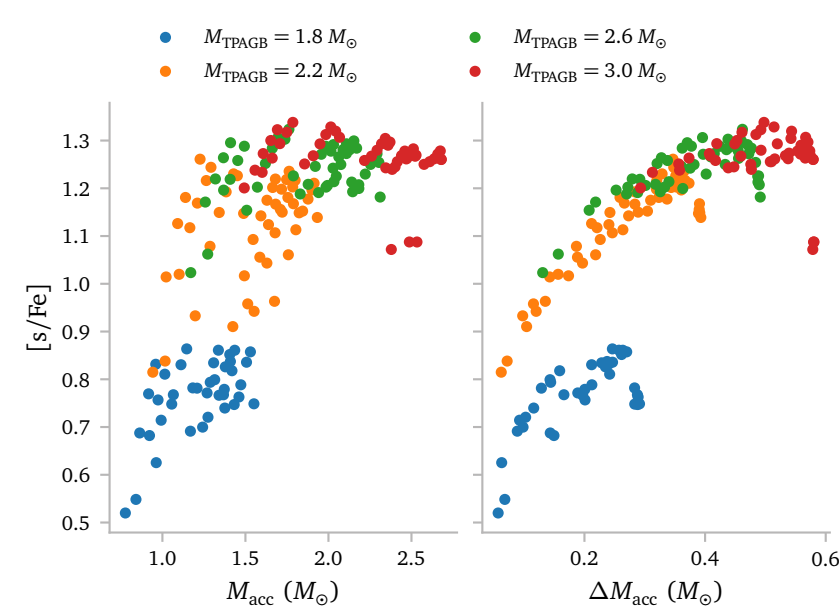
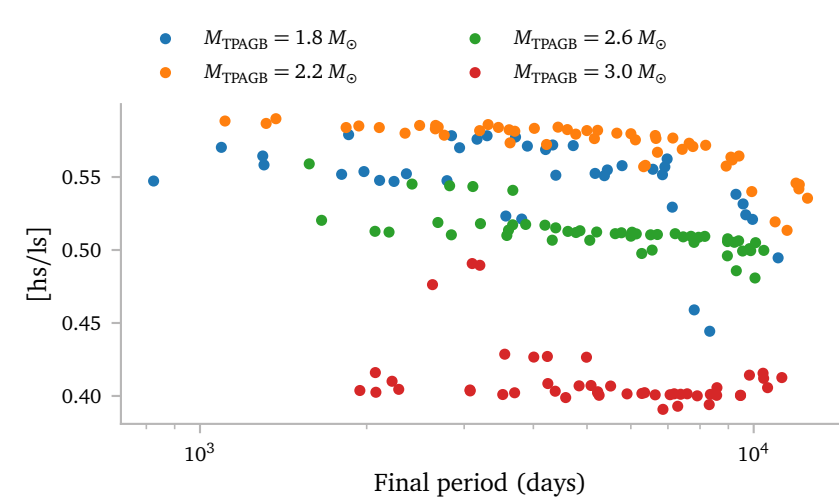
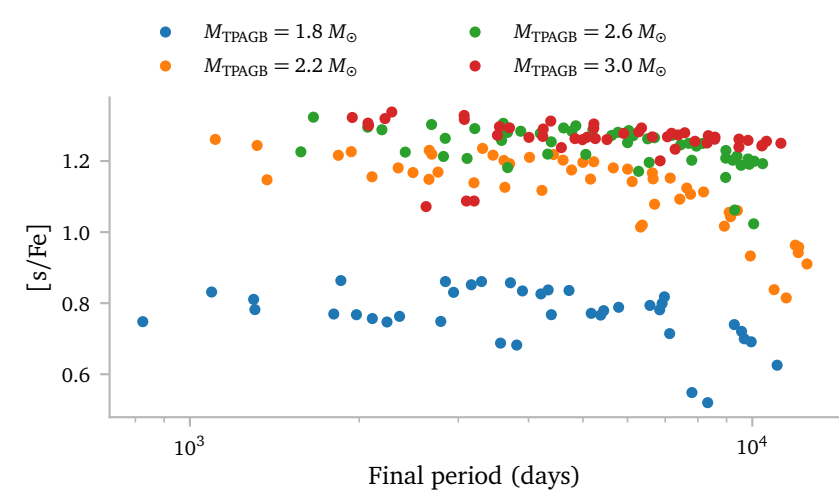
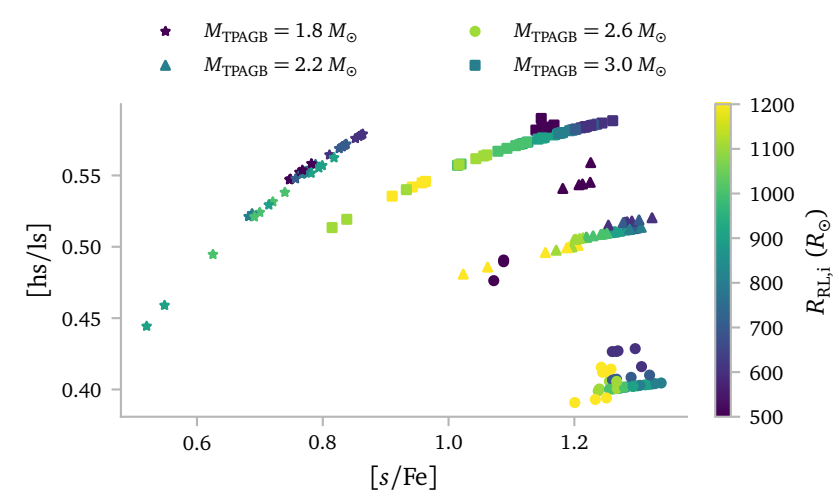
I think I have the most important features in the difference plots. I do not really know where the small bump in the core composition comes from though.

c. $[X/Fe]$ abundances



All of these models have $[s/Fe] \gg 0.25$ and would be classified as s -process enhanced star.

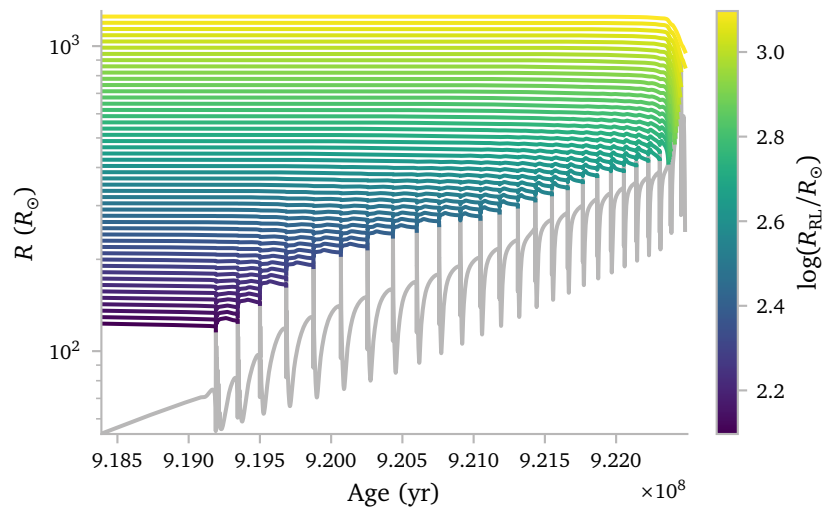
Clearly the lowest mass star has the lowest $[s/Fe]$, which makes sense.



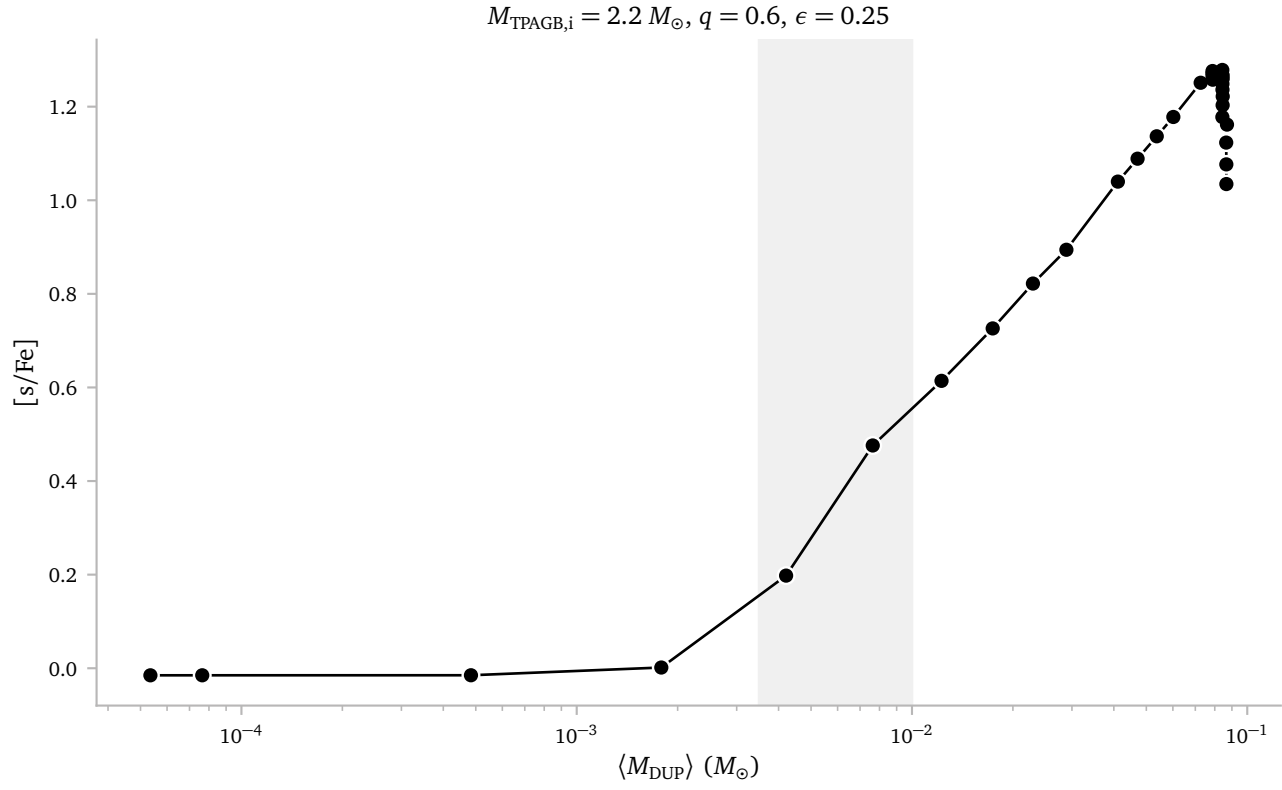
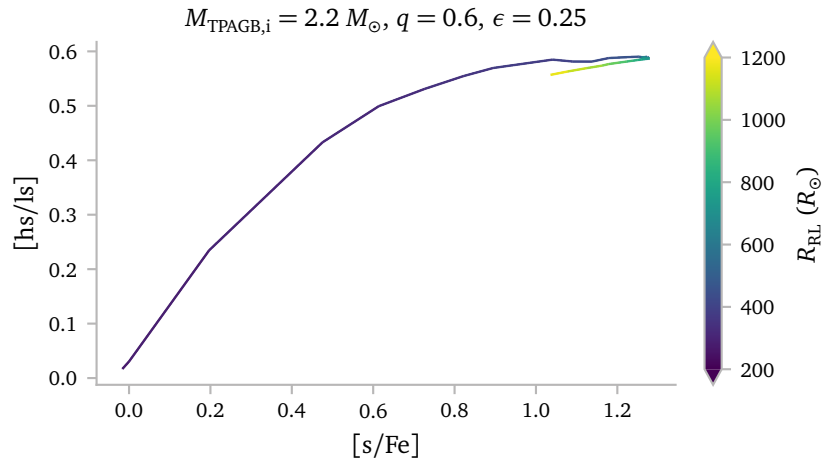
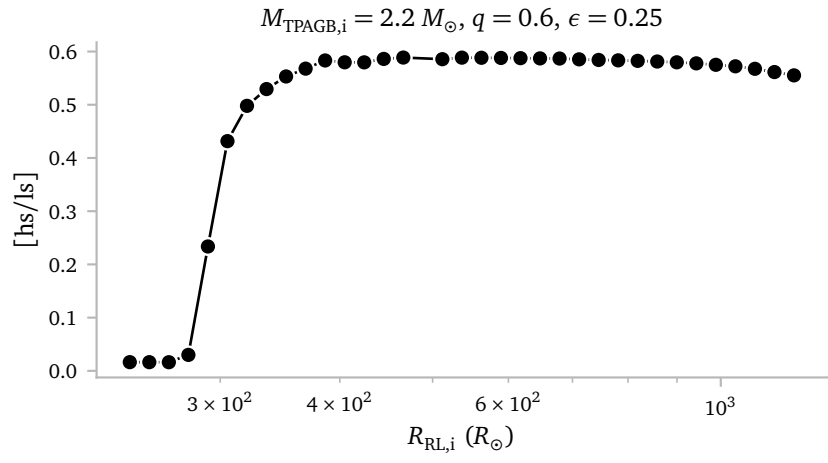
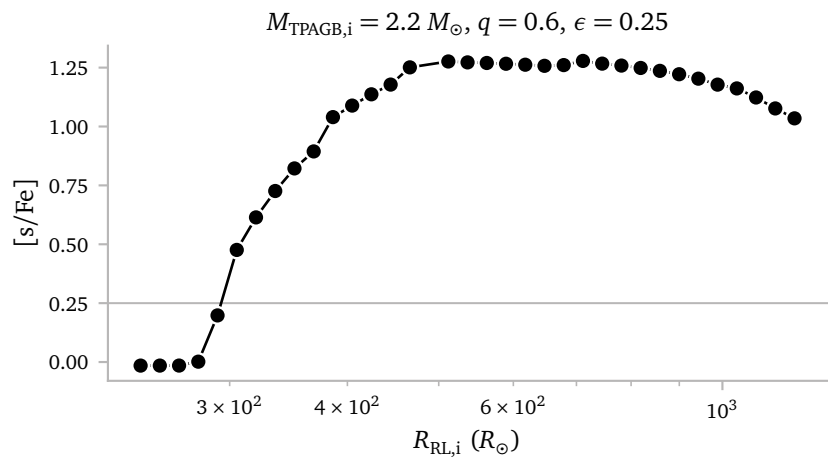
d. Broad radius grid

I think it is interesting to look into the effects of dredge up on the final abundances of the accretor. I want to simulate a MESA model for (nearly) every thermal pulse. This should give me a good idea of the effects of early RLOF interactions on the final abundances.

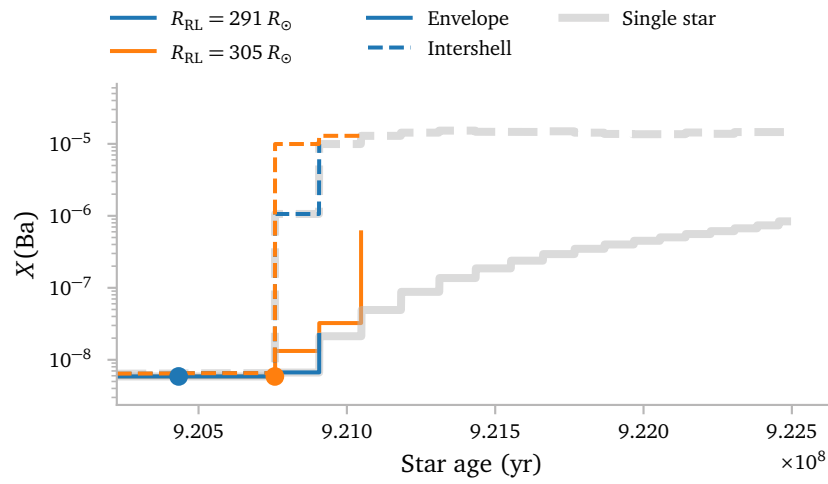
Below, I show the simple binary evolutions on which I based my MESA grid.



Nearly every pulse should have a binary model. The last few models have multiple, because here the radius increases a lot.



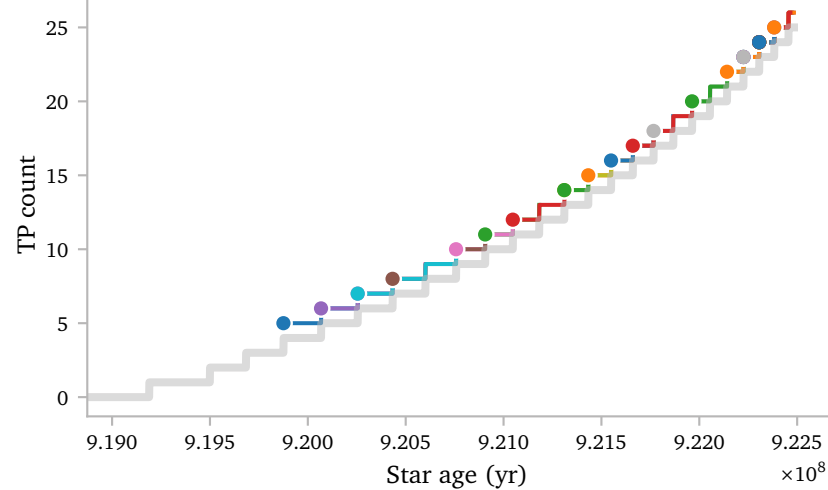
Let me take a look at the time evolution of the intershell and envelope enrichment of the models in the gray band.



It looks like the 305 $R_{\odot}$ model does not follow the single star model in terms of enrichment. This is obviously wrong.

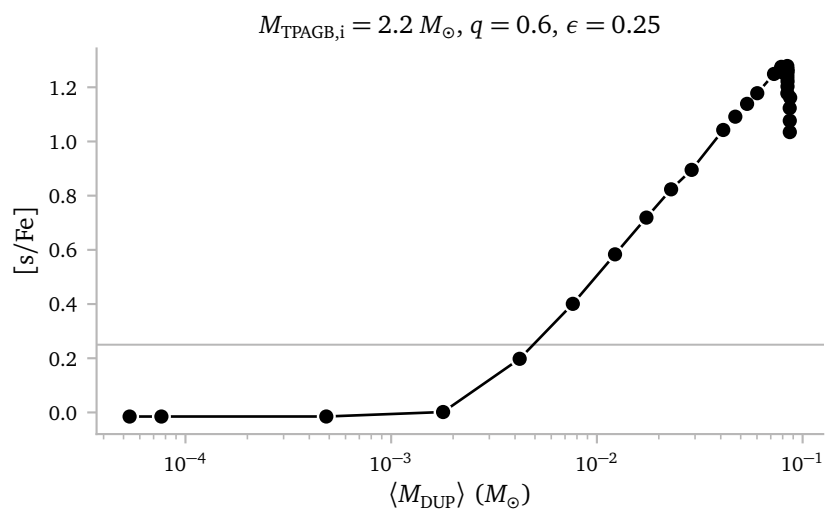
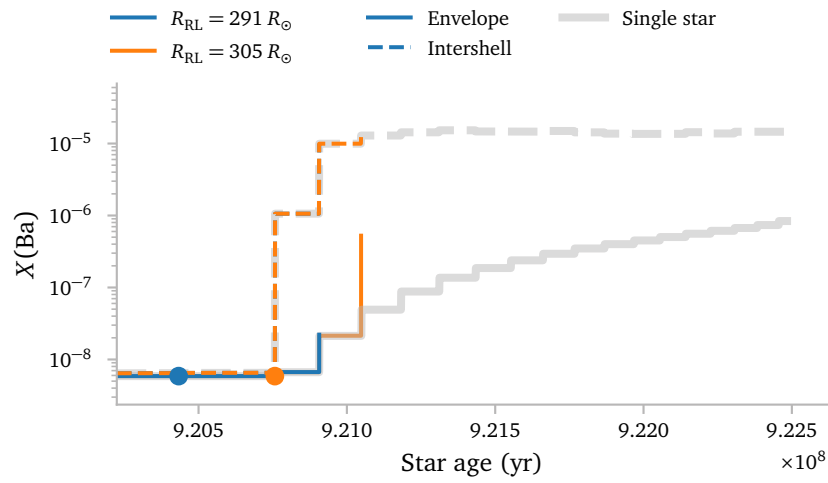
The difficulty lies in finding *why* the model does not follow the single star.

It took me some time to find the cause, but I the plot below shows the problem.



The binary models start with the wrong thermal pulse count. They are offset by exactly 1, which makes sense since the MESA TPAGB routine starts a binary with TP 1, not 0. If I then add the MESA count to the single star count at the starting time of the binary, we get a mismatch. Towards the end of the evolution, this mismatch does not significantly affect envelope abundances anymore. Therefore, the accreted abundances will also not be affected much. However, when dredge up is just starting we can get quite large differences. Luckily the fix is quite easy.

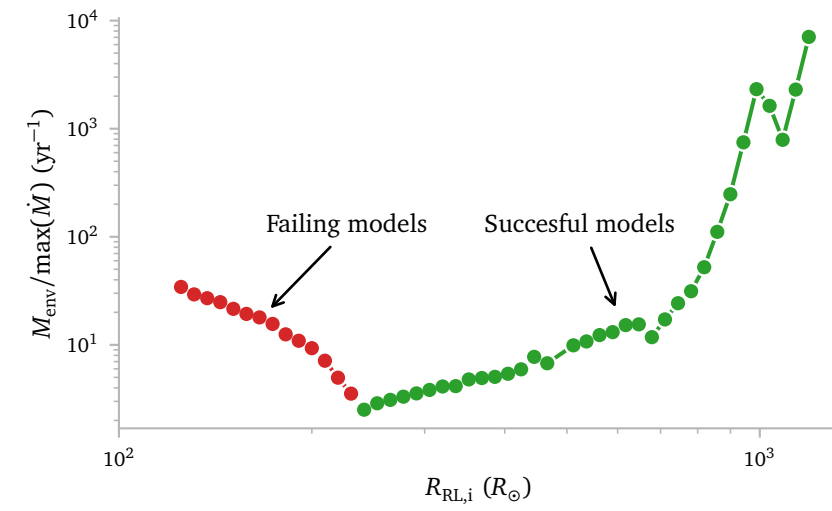
```
from:
self.tp_count[self.simple_end_idx :] =
    self.model.TP_count[:,sampling] +
    simple.TP_count[self.simple_end_idx]
to:
self.tp_count[self.simple_end_idx :] =
    self.model.TP_count[:,sampling]
    -self.model.TP_count[0] +
    simple.TP_count[self.simple_end_idx]
```



The effects are really only clearly visible for the 305 $R_{\odot}$ model, also in the other graphs. I will not recreate these. The general trends still hold.

e. Failing models

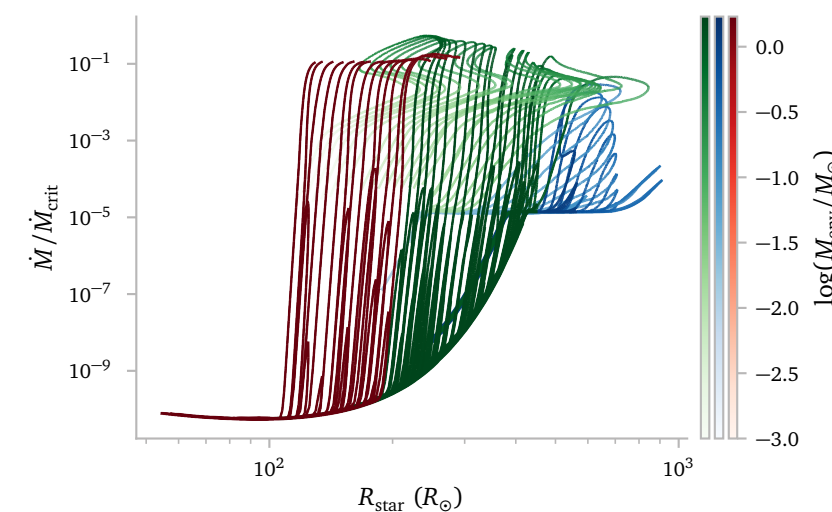
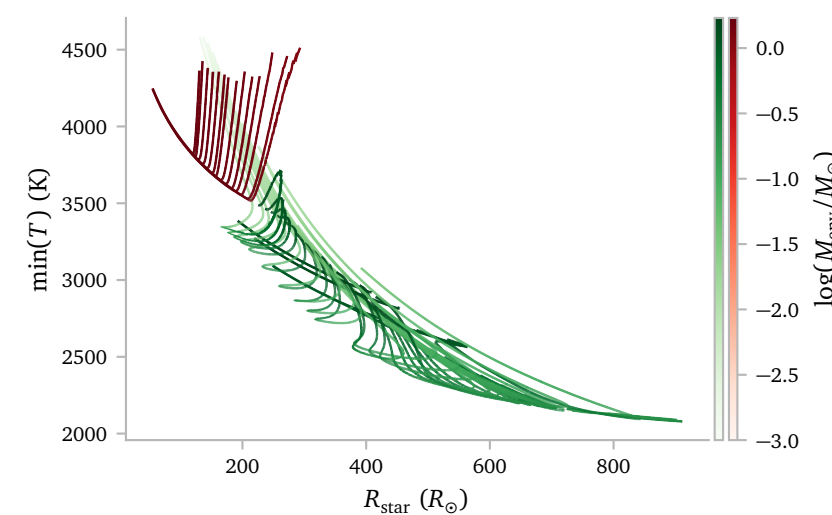
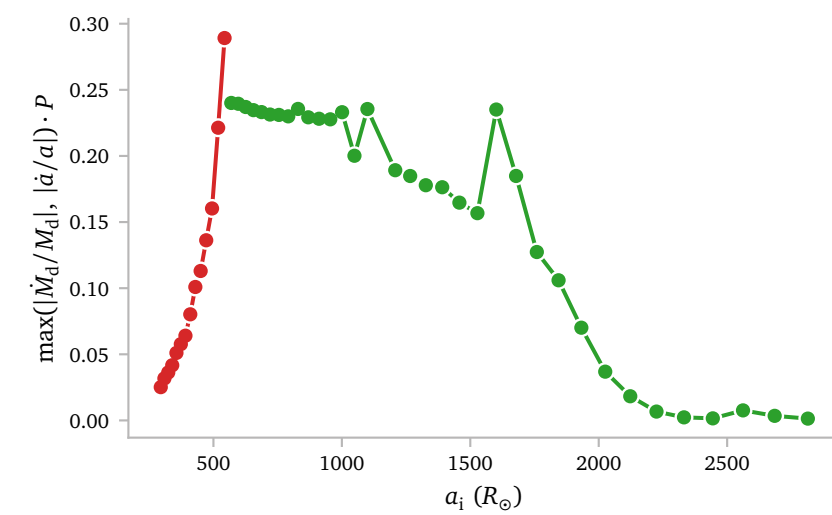
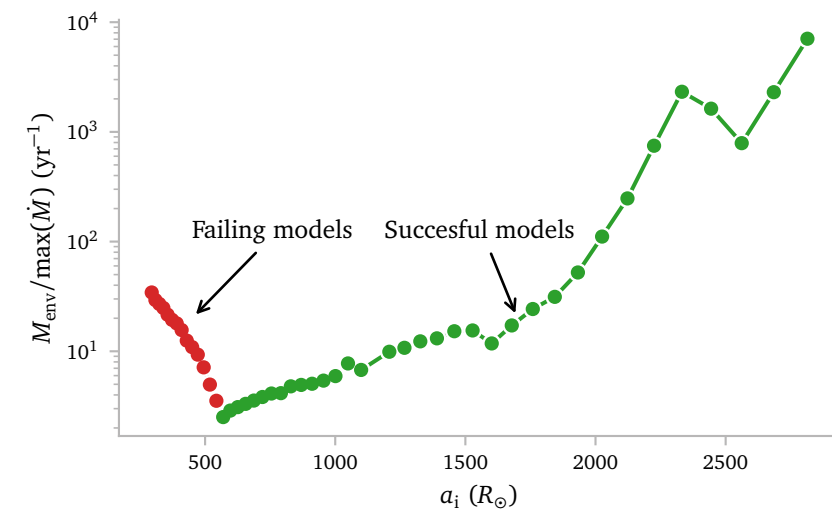
Unfortunately, quite a large portion of my grid to run until the full envelope had been expelled. These models used my automatic restarting script, but they all hit a barrier during a similar part of their evolution which they are unable to move past.



The failing models are *all* smaller than some critical initial Roche lobe radius. In this case, since we have constant q , this is very closely related to the initial binary separation as well, which I plot below.

The vertical axis is a measure for the timescale of envelope stripping at the moment of most extreme mass transfer.

For the failing models however, they *probably* have not reached the mass transfer rate that they would get if they continued their evolution.



References

- Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. (2009) *The Chemical Composition of the Sun*. *ARA&A***47**, 481.
- Karakas, A. I. (2014) *Helium enrichment and carbon-star production in metal-rich populations*. *MNRAS***445**, 347.
- Karakas, A. I. & Lugaro, M. (2016) *Stellar Yields from Metal-rich Asymptotic Giant Branch Models*. *ApJ***825**, 26.